Quality-Aware Preprocessing Selection for Blood Cell Detection: An Empirical Study

An Experimental Evaluation of QAPS-YOLO on a Blood Smear Dataset

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Research prototype — not for clinical use

Abstract

Blood cell detection from smear images is sensitive to acquisition quality. This paper evaluates a Quality-Aware Preprocessing Selection (QAPS) strategy that adaptively applies denoising, gamma correction, and CLAHE based on measured image quality, paired with a YOLO detector. Using a held-out test set of 36 images (471 ground-truth objects), we compare baseline YOLO against QAPS-YOLO at an operating confidence of 0.50. QAPS-YOLO achieves macro F1 of 0.8451 versus 0.8364 for the baseline, a difference that is not statistically significant (paired permutation p = 0.499). Ablation shows that only CLAHE is active on this dataset and slightly reduces F1, while denoising and gamma correction are inactive. Quality-stratified analysis reveals limited quality diversity in the test set. We conclude that adaptive preprocessing must be validated on larger, more diverse data before claiming robustness benefits.

Keywords: blood cell detection, YOLO, quality-aware preprocessing, QAPS, ablation, statistical significance, medical imaging.

1. Introduction

Automated blood cell detection can reduce manual effort in laboratory analysis, but image quality varies with microscope, staining, and illumination. Fixed preprocessing pipelines may improve some images while degrading others. This work studies Quality-Aware Preprocessing Selection (QAPS), which measures image quality and conditionally applies denoising, gamma correction, and CLAHE, then runs YOLO detection. We ask: (RQ1) Does QAPS improve detection over baseline YOLO? (RQ2) Which QAPS components contribute? (RQ3) Is any improvement statistically significant? (RQ4) Does performance vary by image quality?

2. Related Work

Object detection in medical images has been widely studied with YOLO-family models. Image enhancement (CLAHE, gamma correction, denoising) is commonly used to improve contrast, but its effect on detection is inconsistent and may introduce artifacts. Adaptive preprocessing selection remains underexplored for blood smear images. This paper contributes an empirical, statistically grounded evaluation of adaptive preprocessing for blood cell detection.

3. Methodology

3.1 Dataset

We use a blood smear dataset with 765 training, 73 validation, and 36 test images. Ground-truth annotations cover three classes: Platelets, RBC, and WBC. The test set contains 471 objects (398 RBC, 37 WBC, 36 Platelets).

3.2 Detection model

A YOLO11n model trained for 100 epochs (early stopping, cosine LR) on the training split is used. Inference uses image size 640, IoU threshold 0.70, and an operating confidence threshold of 0.50 (selected by F1 tuning on the validation set).

3.3 QAPS

QAPS measures blur (Laplacian variance), brightness/contrast (mean/std of grayscale), and noise (MAD-based residual). It then applies, in order: (1) median denoising if noise >= 8.0; (2) gamma correction if brightness is outside [70, 190] (target 120); (3) CLAHE if contrast < 35. Statistics are recomputed between

steps. Both branches use identical YOLO settings.

3.4 Evaluation protocol

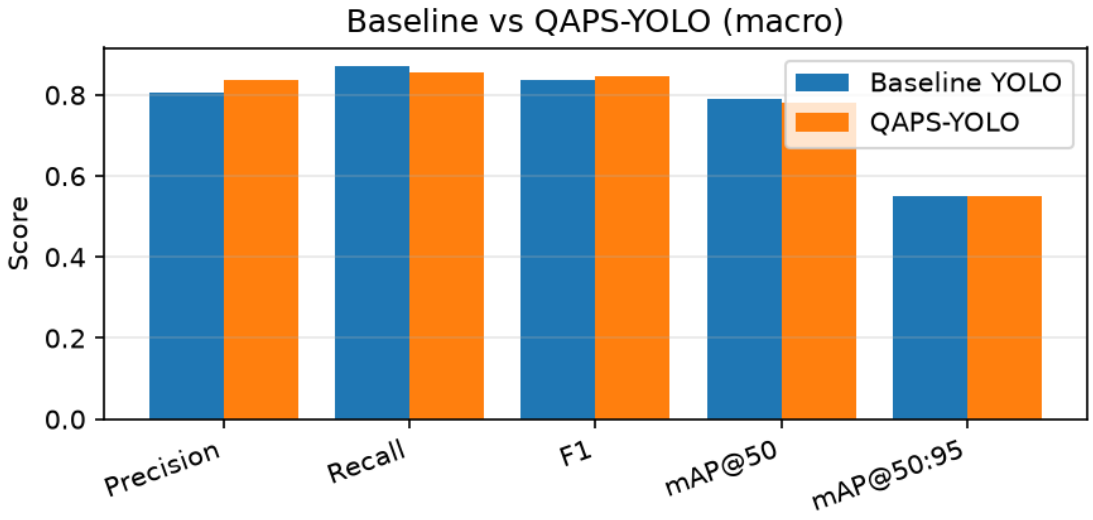
We compute precision, recall, F1, mAP@50, and mAP@50:95 using greedy one-to-one matching at IoU >= 0.5. Paired image-level bootstrap confidence intervals and Wilcoxon/permutation tests assess significance. Ablation disables individual QAPS components. Quality-stratified analysis groups images by quality bands.

4. Results

4.1 Overall comparison

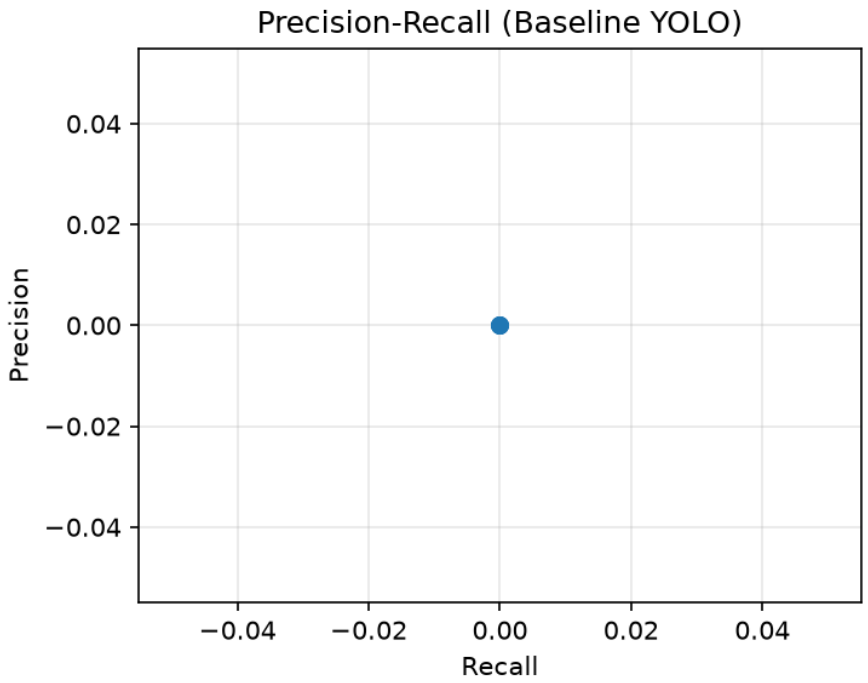
Table 1 summarizes macro-averaged metrics. QAPS-YOLO shows a small F1 improvement (0.8451 vs 0.8364) that is not statistically significant.

Metric	Baseline YOLO	QAPS-YOLO	Delta
Precision	0.8067	0.8361	+0.0294
Recall	0.8727	0.8551	-0.0176
F1	0.8364	0.8451	+0.0088
mAP@50	0.7907	0.7817	-0.0090
mAP@50:95	0.5508	0.5491	-0.0017



4.2 Precision-Recall curve

Precision-recall curve for baseline YOLO across confidence thresholds.



4.3 Per-class comparison

Class	Method	P	R	F1	AP50
Platelets	Baseline YOLO	0.711	0.889	0.790	0.767
RBC	Baseline YOLO	0.736	0.756	0.746	0.659
WBC	Baseline YOLO	0.973	0.973	0.973	0.946
Platelets	QAPS-YOLO	0.795	0.861	0.827	0.744
RBC	QAPS-YOLO	0.740	0.731	0.736	0.641
WBC	QAPS-YOLO	0.973	0.973	0.973	0.960

4.3 Confusion summary (TP/FP/FN)

Class	Method	TP	FP	FN
Platelets	Baseline YOLO	32	13	4
RBC	Baseline YOLO	301	108	97
WBC	Baseline YOLO	36	1	1
Platelets	QAPS-YOLO	31	8	5
RBC	QAPS-YOLO	291	102	107
WBC	QAPS-YOLO	36	1	1

4.3 Ablation study

Table 2 shows the effect of disabling individual QAPS components. Denoising and gamma correction produce identical results to full QAPS (inactive on this data), while disabling CLAHE reverts to baseline performance.

Variant	Precision	Recall	F1
Baseline	0.7392	0.8052	0.7450
QAPS full	0.7494	0.7850	0.7403

QAPS no denoise	0.7494	0.7850	0.7403
QAPS no gamma	0.7494	0.7850	0.7403
QAPS no CLAHE	0.7392	0.8052	0.7450

4.4 Quality-stratified analysis

All 36 test images fall into the same quality band (blurry, normal brightness, clean), so no per-group differences can be reported. This indicates limited quality diversity in the test set.

4.5 Statistical significance

Paired image-level analysis: mean F1 difference = -0.0029 (95% CI [-0.0102, 0.0030]), permutation p = 0.4986, Wilcoxon p = 0.5850, Cohen's dz = -0.137. After Holm correction, no comparison is significant at alpha = 0.05. 11 images improved, 10 unchanged, 15 worsened.

4.6 Operating threshold tuning

Tuning the confidence threshold on the validation set selected 0.50 as the best F1 operating point (F1 = 0.7694), improving precision at a modest recall cost.

5. Discussion

QAPS does not provide a statistically significant benefit on this dataset. Ablation reveals that denoising and gamma correction are inactive, and CLAHE slightly reduces F1. This suggests that adaptive preprocessing selection, as implemented, is not yet beneficial for detection on this data. The lack of quality diversity in the test set limits robustness conclusions. Threshold tuning improves the operating point, indicating that confidence selection matters more than preprocessing here.

6. Limitations

Small test set (36 images), class imbalance (398 RBC vs 36 platelets vs 37 WBC), limited quality diversity, no external/multi-site validation, and no calibration to physical cell concentrations. Counts are detections per image, not clinical measurements.

7. Conclusion

We evaluated QAPS-YOLO for blood cell detection and found no statistically significant improvement over baseline YOLO. Ablation showed CLAHE as the only active component, slightly reducing F1. Future work should use larger, more diverse datasets, learnable preprocessing policies, and clinical calibration.

References

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